STRUCTURE, INVARIANTS AND OBSTRUCTIONS IN COLLATZ DYNAMICS: A 2-ADIC OBSTRUCTION TO MODULAR LYAPUNOV FUNCTIONS AND EXACT WASSERSTEIN CONTRACTION COEFFICIENTS

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ABSTRACT. We study the accelerated Collatz map $T(n) = n/2$ (n even), $T(n) = (3n+1)/2$ (n odd), and its affine siblings $3n+d$. A natural strategy for proving global convergence is to construct a modular Lyapunov function, correcting the macroscopic logarithmic descent with periodic weights.

Our main result is a nonexistence theorem for this approach: for every $j \geq 1$ there is no function $V(n) = \log_2 n + w(n \bmod 2^j)$, with $w : \mathbb{Z}/2^j\mathbb{Z} \to \mathbb{R}$ arbitrary, that decreases strictly along the odd orbits of T in $\mathbb{Z}_+$. The obstruction is structural: the 2-adic fixed point $-1 \in \mathbb{Z}_2$ projects, in every modulus 2^j, onto a residue class carrying a self-loop of logarithmic gain $\log_2 3 - 1 > 0$ that no modular potential can compensate. The mechanism is robust to finite sets of exceptions, to non-strict decrease, and to conditions imposed on blocks of k iterations.

We complement this with two contraction theorems for the transfer structure of the map: the Syracuse Markov chain on $\mathbb{Z}_3$ has Dobrushin–Wasserstein coefficient at most $1/3$ (each inverse branch scales the 3-adic metric by exactly $1/3$), yielding a unique invariant measure and geometric W_1-equidistribution; and the dual chain on $\mathbb{Z}_2$ has coefficient exactly $1/2$ and averages every level-k cylinder function to its exact Haar mean in k steps.

All computational claims are produced by exact algorithms — rational arithmetic, arbitrary-precision integers, and integer certificates for the continued fraction of $\log_2 3$ — implemented in a small, dependency-free Python library. Every number quoted in the paper is regenerated by a single reproduction script. The computations instantiate, but never replace, the proofs: numerical sweeps are labelled as verifications, with limits stated explicitly.

1. Introduction

Let $T : \mathbb{Z} \to \mathbb{Z}$ be the accelerated Collatz map,

$$T(n) = \begin{cases} n/2, & n \text{ even}, \\ (3n+1)/2, & n \text{ odd}. \end{cases}$$

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The Collatz conjecture asserts that every $n \geq 1$ reaches the cycle $\{1, 2\}$ under iteration of T. The problem has an extensive literature, surveyed by Lagarias [6, 7]; the strongest partial results include Terras's theorem that almost every n has finite stopping time [10], the density bounds of Krasikov and Lagarias [5], lower bounds on the length of hypothetical nontrivial cycles by Eliahou [2] and Simons–de Weger [8], Tao's theorem that almost all orbits attain almost bounded values [9], and the computational verification of convergence for all $n \leq 2^{68}$ by Barina [1]. Functional-analytic and Markov-chain approaches to the $3n+1$ map, close in spirit to our Section 4, are developed systematically by Wirsching [12].

This paper has two aims. The first is to prove a nonexistence theorem for a natural class of candidate Lyapunov functions (Section 3) and two exact contraction theorems for the transfer structure of the map on the 2-adic and 3-adic completions (Section 4). The second is methodological: we describe a small computational laboratory, built exclusively on exact arithmetic, whose outputs instantiate the theory — enumerating cycles as solutions of a Diophantine equation, certifying continued-fraction data by integer comparisons, and computing Wasserstein contraction coefficients as exact rationals (Sections 5–7).

1.1. **Positioning and motivation.** The search for a global convergence certificate for the Collatz conjecture naturally leads to the study of Lyapunov functions. Because the length-k parity vector of an orbit is entirely determined by the residue modulo 2^k [10], modular Lyapunov functions—which perturb the macroscopic logarithmic descent with finite-dimensional periodic corrections—are widely considered a highly promising strategy. Much of the heuristic and rigorous work on the conjecture, especially algebraic attempts to bound trajectories based on parity patterns, implicitly relies on the hope that some sufficiently deep modular correction can enforce strict monotonicity. Our nonexistence theorem rules out this entire class of functions. We show that the barrier is not merely that the correct weights have not yet been found, but rather that a fundamental topological obstruction unconditionally prevents any such function from existing.

In parallel with pointwise dynamics, the macroscopic behavior of the Collatz map is often studied via Markov chains and transfer operators [12, 9]. While finite-dimensional projections (such as the Syracuse chain modulo 3^k) are known to exhibit rapid mixing, their exact spectral structure in ℓ^2 is trivial. We shift the analysis to the Wasserstein metric on the p-adic completions $\mathbb{Z}_2$ and $\mathbb{Z}_3$. Our geometric contraction results provide exact, dimension-free Dobrushin–Wasserstein coefficients for these operators. These exact bounds provide simpler, self-contained proofs of unique ergodicity and equidistribution that bypass traditional functional-analytic machinery, leveraging the strict ultrametric geometry of the inverse branches.

1.2. **Main contributions.**

(1) **Contribution A: Complete modular obstruction theorem** (Theorem 3.2). Mathematically, we prove that for every $j \geq 1$ there is no strictly decreasing function of the form $V(n) = \log_2 n + w(n \bmod 2^j)$ along odd orbits in $\mathbb{Z}_+$. The proof introduces a novel structural observation: the fixed point $T(-1) = -1$ in $\mathbb{Z}_2$ projects onto the residue $-1 \bmod 2^j$, and the family $n_m = 2^{j+1}m - 1$ makes the modular potential cancel exactly while the logarithmic term grows. Four robustness corollaries (Corollary 3.4) show that the usual relaxations — finitely many exceptions, non-strict decrease, decrease along blocks of k iterations, restricted domains — do not evade the obstruction.

(2) **Contribution B: Exact Dobrushin coefficient on the dual chains** (Theorems 4.3 and 4.5). Mathematically, we establish uniform contraction of the transfer operators in the Wasserstein metric. The Syracuse Markov chain on $\mathbb{Z}_3$ is a uniformly contractive iterated function system: every inverse branch scales the 3-adic metric by exactly 1/3, yielding a Dobrushin–Wasserstein coefficient of at most 1/3. Dually, the backward chain on $\mathbb{Z}_2$ has a coefficient of exactly 1/2. This provides a simpler, geometrically explicit mechanism for Haar averaging, mapping every level-k cylinder function to its exact Haar average in exactly k steps.

(3) **Contribution C: Exact reproducible computational framework** (Sections 5–7). Algorithmically and technically, we provide a computational laboratory built entirely on exact rational and arbitrary-precision integer arithmetic. We implement cycle enumeration via the closed form $n = b(p)/(2^L - 3^k)$; Diophantine cycle exclusion via certified integer convergents of $\log_2 3$ [2, 8]; identification of the obstruction at every finite level by Karp's maximum-mean-cycle algorithm [3]; and exact rational evaluation of Wasserstein coefficients.

1.3. **Intuition and phenomenology.** Before proceeding to the formal proofs, it is helpful to build a mental picture of the obstruction. What should the reader picture?

- **Why modular corrections are attractive:** The operation $T(n)$ divides by 2 when n is even and multiplies by roughly 3/2 when n is odd. The exact sequence of operations depends purely on the dyadic valuation of the trajectory, which is locally determined by the residue of n modulo 2^j. A modular potential $w(n \bmod 2^j)$ is therefore the natural candidate to "smooth out" the local fluctuations of the orbit.
- **Why the residue graph matters:** We can view the transitions modulo 2^j as a directed graph (Section 5.3). The edges carry a "cost" corresponding to the logarithmic gain or loss of the operation. A valid Lyapunov function requires finding a potential (node

weights) that absorbs these edge costs, rendering every path strictly decreasing.

- **Why the fixed point -1 is unavoidable:** The Collatz map extends continuously to the 2-adic integers $\mathbb{Z}_2$, where $T(-1) = -1$ is a fixed point. When projected down to the finite ring $\mathbb{Z}/2^j\mathbb{Z}$, this fixed point appears as the residue class $-1 \bmod 2^j$.
- **Why the self-loop creates an obstruction:** The residue $-1 \bmod 2^j$ invariably loops to itself. If one approaches -1 along the positive integers via the family $n_m = 2^{j+1}m - 1$, the modular potential perfectly cancels out (since start and end residues are identical), but the true logarithmic value experiences the unmitigated growth of the $3n + 1$ operation (a gain of roughly $\log_2 3 - 1 > 0$). This localized "heat source" breaks any attempt to construct a globally decreasing potential.

The scientific significance of Theorem 3.2 is that it identifies a structural barrier to a highly natural proof strategy. Because the parity vector of length j is determined precisely by the residue modulo 2^j, a vast amount of heuristic and rigorous work implicitly seeks a convergence certificate (a Lyapunov function) whose corrections depend only on finite-dimensional modular data. Our theorem establishes that all such attempts face an insurmountable topological obstruction: the difficulty is not that the correct modular weights have not yet been found, but that the 2-adic fixed point -1 projects a strictly positive logarithmic gain into every finite quotient, unconditionally breaking any such function even when the domain is strictly restricted to the positive integers.

1.4. **Theorems versus verifications.** We maintain a strict separation between the two kinds of statement in this paper.

- *Theorems* (Sections 3–5) are proved analytically; none of their proofs depends on a computation.
- *Verifications* (Section 7) are outputs of the algorithms of Section 5, reported with their exact sweep limits. They serve two purposes: validating the implementations against known ground truth (the negative cycles of $3n+1$, the nontrivial cycles of $3n-1$ and $3n+5$), and instantiating the theorems at finite levels (the Karp analysis locating $-1 \bmod 2^j$, the coefficients τ_k increasing towards the bound $1/3$).

In particular, our convergence sieve ($n \leq 2 \cdot 10^5$) makes no claim of novelty — the published record is 2^{68} [1] — and is used only as a self-contained hypothesis for the exclusion bounds of Section 7.3, which we then also evaluate at the literature's limit.

1.5. **Reproducibility.** All algorithms are implemented in a pure-Python package `collatz` (no third-party dependencies) accompanying this paper, with a test suite and a single script, `reproduce_paper_results.py`, that regenerates and checks every number quoted here (its sections R1–R8 are cross-referenced in Section 7). The arithmetic model, and why no floating-point issue can affect any certified claim, is specified in Section 6.

2. Notation and preliminaries

$\mathbb{Z}_+$ denotes the positive integers. For a prime p, $\mathbb{Z}_p$ is the ring of p-adic integers, v_p the p-adic valuation and $|x|_p = p^{-v_p(x)}$ the p-adic absolute value; $(\mathbb{Z}_p, |\cdot|_p)$ is a compact ultrametric space. $\nu_2(n)$ is the 2-adic valuation of an integer n. For $x \in \mathbb{R}$, $\|x\|$ denotes the distance from x to the nearest integer. Throughout, $\alpha = \log_2 3$.

Definition 2.1 (Collatz maps)**.** For an integer parameter d odd, the *accelerated map* of the $3n + d$ system is

$$T_d(n) = \begin{cases} n/2, & n \text{ even}, \\ (3n+d)/2, & n \text{ odd}, \end{cases}$$

and we write $T = T_1$. The *Syracuse map* S acts on odd integers by $S(n) = (3n+1)/2^{\nu_2(3n+1)}$. The *parity vector* of length k of n is $\big(n \bmod 2,\, T(n) \bmod 2,\, \ldots,\, T^{k-1}(n) \bmod 2\big) \in \{0,1\}^k$.

We will use Terras's structural theorem: the length-k parity vector, as a function of $n \bmod 2^k$, is a bijection of $\mathbb{Z}/2^k\mathbb{Z}$ onto $\{0,1\}^k$, and T extends to $\mathbb{Z}_2$ where it is topologically conjugate to the one-sided shift [10, 6].

Definition 2.2 (Wasserstein distance and contraction coefficient)**.** Let (X, d) be a compact metric space and μ, ν Borel probability measures on X. The Kantorovich–Rubinstein distance is

$$W_1(\mu,\nu) \;=\; \inf_{\gamma\in\Gamma(\mu,\nu)} \int_{X\times X} d(x,y)\,\mathrm{d}\gamma(x,y),$$

the infimum over couplings of μ and ν [11, Ch. 6]. For a Markov kernel P on X, the *Dobrushin–Wasserstein contraction coefficient* is

$$\tau(P) \;=\; \sup_{x\neq y} \frac{W_1\big(P(x,\cdot),\, P(y,\cdot)\big)}{d(x,y)}.$$

The standard consequences we use are collected in the next lemma.

Lemma 2.3. *Let P be a Markov kernel on a compact (hence complete separable) metric space (X, d) with $\tau = \tau(P) < 1$. Then:*

1. *$W_1(\mu P, \nu P) \le \tau\, W_1(\mu,\nu)$ for all Borel probability measures μ, ν;*
2. *P has a unique invariant Borel probability measure π, and for every μ, the sequence μP^n converges to π in the Wasserstein metric, with $W_1(\mu P^n, \pi) \le \tau^n\, W_1(\mu, \pi)$;*
3. *the Koopman operator $Uf(x) = \int f\,\mathrm{d}P(x,\cdot)$ satisfies $\mathrm{Lip}(Uf) \le \tau\,\mathrm{Lip}(f)$; hence on the quotient of the Lipschitz space by the constants, the spectral radius of U is at most τ, i.e. $\mathrm{spec}\big(U|_{\mathrm{Lip}}\big) \subseteq \{1\} \cup \{z : |z| \le \tau\}$.*

Proof. (1) Given a coupling γ of (μ,ν) and, for each pair (x,y), a coupling $\gamma_{x,y}$ of $(P(x,\cdot),P(y,\cdot))$ with cost at most $\tau\, d(x,y)+\varepsilon$, the mixture $\int \gamma_{x,y}\,\mathrm{d}\gamma$ couples μP and νP with cost at most $\tau \int d\,\mathrm{d}\gamma + \varepsilon$; take infima. (2) X compact implies the space of Borel probability measures with W_1 is a complete metric space [11, Thm. 6.18]; by (1) the map $\mu \mapsto \mu P$ is a τ-contraction, so the Banach fixed point theorem applies. (3) For f Lipschitz and $x \neq y$, Kantorovich–Rubinstein duality [11, Ch. 5] gives $|Uf(x) - Uf(y)| \leq \mathrm{Lip}(f)\, W_1(P(x,\cdot),P(y,\cdot)) \leq \mathrm{Lip}(f)\,\tau\, d(x,y)$. Iterating, $\mathrm{Lip}(U^n f) \leq \tau^n \,\mathrm{Lip}(f)$, so on Lip /constants the operator norm of U^n is at most τ^n. ☐

3. The modular obstruction to Lyapunov functions

A natural strategy for proving global convergence is to exhibit a Lyapunov function combining the macroscopic logarithmic trend of the orbit with periodic local corrections.

Definition 3.1 (Modular Lyapunov function)**.** Fix $j \geq 1$. A *modular Lyapunov function of level j* is a function

$$V(n) \;=\; \log_2 n + w(n \bmod 2^j), \qquad w : \mathbb{Z}/2^j\mathbb{Z} \to \mathbb{R} \text{ arbitrary},$$

such that $V(T(n)) < V(n)$ for every odd integer $n > 0$.

Since w has finite domain, it is bounded, and on the integers it acts as a periodic potential of period 2^j. If such a V existed, no orbit could diverge (the $\log_2$ term would be forced below any bound along the infinitely many odd steps of a divergent orbit while even steps also decrease V, since $V(n/2) = V(n) - 1 + w(\frac{n}{2} \bmod 2^j) - w(n \bmod 2^j)$ is controlled by the same potential — but we will not need this motivation). Our main result is that the class is empty.

Theorem 3.2 (Modular obstruction)**.** ***Assumptions:*** *Let $j \geq 1$ be an integer. Let $V(n) = \log_2 n + w(n \bmod 2^j)$ be a candidate modular Lyapunov function, where $w : \mathbb{Z}/2^j\mathbb{Z} \to \mathbb{R}$ is an arbitrary function.*

Conclusion: *The function V cannot be strictly decreasing along the odd orbits of T in $\mathbb{Z}_+$. Specifically, the projection of the 2-adic fixed point $-1 \in \mathbb{Z}_2$ onto $\mathbb{Z}/2^j\mathbb{Z}$ induces, on the residue class $-1 \bmod 2^j$, a growth loop of logarithmic gain $\log_2 3 - 1 > 0$ that no potential w can compensate.*

Interpretation: This theorem means that we cannot prove the Collatz conjecture merely by adding a periodic, finite-dimensional correction to the macroscopic logarithmic descent. It rules out a large implicit class of bounds derived purely from modular constraints. However, it still leaves open the possibility that non-periodic corrections, or corrections depending on infinitely many residues, could form a valid Lyapunov function.

Scope and limitations: The impossibility result applies explicitly to modular corrections $w(n \bmod 2^j)$. It is natural to ask whether more complex corrections could evade this.

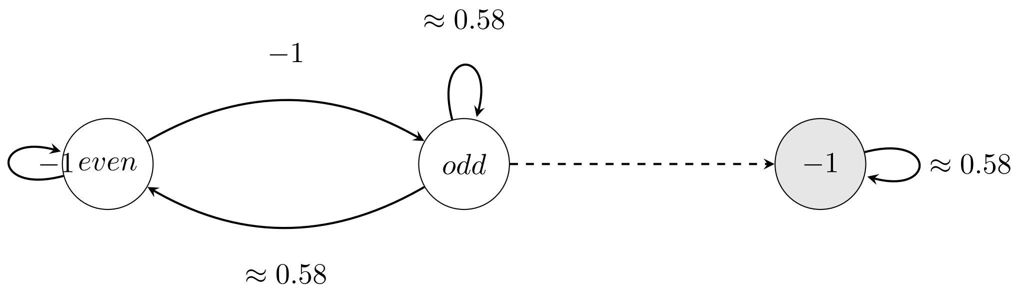

FIGURE 1. A schematic visualization of the residue graph and the obstruction self-loop at the class containing -1. The edges represent transitions modulo 2^j, carrying logarithmic costs. The unavoidable self-loop at $-1 \bmod 2^j$ accumulates a strictly positive cost $\log_2 3-1 \approx 0.58$ at each step, preventing any globally decreasing potential.

- *Polynomial or nonlinear periodic corrections:* Any correction that ultimately factors through a finite quotient $\mathbb{Z}/2^j\mathbb{Z}$ is subject to this obstruction, regardless of its algebraic form.
- *Nonlocal corrections:* A potential depending on the full forward trajectory, or on infinitely many 2-adic digits, is *not* ruled out by this theorem.
- *Adaptive weights or probabilistic Lyapunov functions:* These fall outside the scope of Theorem 3.2. It is currently unknown whether a probabilistic Lyapunov certificate (one that decreases only in expectation over appropriate random choices) could exist.

Proof. Suppose, for contradiction, that there is $w : \mathbb{Z}/2^j\mathbb{Z} \to \mathbb{R}$ such that for every odd $n > 0$,

$$w\big(T(n) \bmod 2^j\big) - w\big(n \bmod 2^j\big) \ < \ -\log_2 \frac{T(n)}{n}. \tag{1}$$

Consider the family $n_m = 2^{j+1}m - 1$ for integers $m \geq 1$. Since $j \geq 1$ we have $n_m \geq 3$, so every n_m is a strictly positive odd integer and lies in the domain where (1) is assumed. The accelerated step evaluates exactly to

$$T(n_m) \ = \ \frac{3(2^{j+1}m-1)+1}{2} \ = \ 3 \cdot 2^j m - 1.$$

Reducing modulo 2^j,

$$n_m = 2^j(2m) - 1 \equiv -1, \quad T(n_m) = 2^j(3m) - 1 \equiv -1 \pmod{2^j}.$$

Substituting n_m into (1), the potential terms cancel identically, leaving, for every $m \geq 1$,

$$0 \ < \ -\log_2 \frac{3 \cdot 2^j m - 1}{2 \cdot 2^j m - 1}.$$

This requires the argument of the logarithm to be strictly less than 1; as both numerator and denominator are positive, that means $3 \cdot 2^j m - 1 < 2 \cdot 2^j m - 1$, i.e. $3 < 2$ — a contradiction. $\square$

Remark 3.3 (The mechanism). In $\mathbb{Z}_2$ the integer -1 is an odd fixed point of the accelerated map: $T(-1) = (3 \cdot (-1) + 1)/2 = -1$. Any function of the residues modulo 2^j is blind to the difference between the positive integers $n_m \equiv -1 \pmod{2^{j+1}}$, which we want to control, and the point $-1 \in \mathbb{Z}_2$ they converge to 2-adically, on which the dynamics genuinely gains $\log_2 \frac{3}{2} = \log_2 3 - 1 > 0$ per step. The family n_m transports this stationary gain into $\mathbb{Z}_+$, where the potential cancels and only the gain remains. The obstruction is thus *structural*: it does not depend on the combinatorics of w, only on the existence of the 2-adic fixed point.

Corollary 3.4 (Robustness)**.** *The obstruction of Theorem 3.2 persists under each of the following relaxations of Definition 3.1.*

(1) Finite exceptions. *If* (1) *is only required outside a finite set $E \subset \mathbb{Z}_+$, no such V exists.*
(2) Non-strict decrease. *If the requirement is weakened to $V(T(n)) \leq V(n)$, no such V exists.*
(3) Blocks of k iterations. *If the requirement is $V(T^k(n)) < V(n)$ for a fixed $k \geq 1$ (for all odd $n > 0$ whose first k accelerated steps are odd, or for all odd $n > 0$ outright), no such V exists.*
(4) Restricted domains. *If the decrease is required only on a subset $D \subseteq \mathbb{Z}_+$ of the odd integers, then either D omits infinitely many terms of some family $n_m = 2^{j+1}m - 1$ — in which case V cannot certify global convergence — or D contains infinitely many of them, and no such V exists.*

Proof. (1) The test family $\{n_m\}_{m \geq 1}$ is infinite, so for any finite E there is m with $n_m \notin E$; the proof of Theorem 3.2 applies verbatim to that m.

(2) With $\leq$ in place of $<$, the final inequality of the proof becomes $3 \cdot 2^j m - 1 \leq 2 \cdot 2^j m - 1$, i.e. $3 \leq 2$, still false.

(3) Parametrize the deeper family $n_m = 2^{j+k}m - 1$, $m \geq 1$. We claim $T^i(n_m) = 3^i \, 2^{j+k-i} m - 1$ for $0 \leq i \leq k$, each of the first k steps being odd. Indeed, by induction: $T^0(n_m) = n_m$; if $T^i(n_m) = 3^i 2^{j+k-i} m - 1$ with $i < k$, then $j + k - i \geq j + 1 \geq 2$, so $3^i 2^{j+k-i} m$ is even and $T^i(n_m)$ is odd, whence

$$T^{i+1}(n_m) = \frac{3\big(3^i 2^{j+k-i} m - 1\big) + 1}{2} = 3^{i+1} 2^{j+k-i-1} m - 1.$$

Hence $T^k(n_m) = 3^k 2^j m - 1 \equiv -1 \pmod{2^j}$ and $n_m \equiv -1 \pmod{2^j}$, so the potentials cancel in the block inequality $w(T^k(n_m) \bmod 2^j) - w(n_m \bmod 2^j) < -\log_2 \frac{T^k(n_m)}{n_m}$, leaving

$$0 < -\log_2 \frac{3^k \, 2^j m - 1}{2^k \, 2^j m - 1},$$

which forces $3^k < 2^k$, false for every $k \geq 1$.

(4) If D contains infinitely many n_m, run the proof of Theorem 3.2 on those; if it omits all but finitely many, then the orbits through the omitted integers are unconstrained by V, which therefore proves nothing about them. $\square$

Remark 3.5 (Relation to the literature). That *some* obstruction to modular Lyapunov functions must exist over all of $\mathbb{Z}$ is unsurprising: the negative integers contain genuine cycles (-1; $-5 \to -7 \to -10$; and the cycle of -17; see Section 7.2), and a strictly decreasing V defined on all of $\mathbb{Z} \setminus \{0\}$ would forbid them. The content of Theorem 3.2 is sharper: the hypothesis is imposed *only on* $\mathbb{Z}_+$, where no cycle other than $\{1, 2\}$ is known, and the failure is nevertheless unconditional, caused by the projection of the single 2-adic point -1 rather than by any actual cycle in the domain of the hypothesis. To the best of our verification, this specific formulation does not seem to appear explicitly in the classical or contemporary literature we consulted [10, 6, 2, 5, 8, 7, 9], and may constitute a novel formulation. We state this merely as a cautious bibliographic observation. Section 5.3 recasts the theorem as the infeasibility of a system of difference constraints and Section 7.4 locates the obstruction computationally at every level $5 \leq j \leq 10$.

4. Wasserstein contraction of the transfer structure

This section proves the two contraction theorems. We first fix the probabilistic model, which is standard [9].

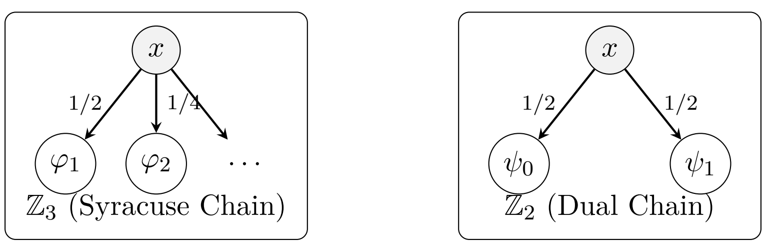

Figure 2. The two transfer models studied in this section. Left: The Syracuse chain on $\mathbb{Z}_3$ uses infinitely many branches $\varphi_a(x) = (3x+1)2^{-a}$ with geometric probabilities, representing the number of powers of 2 divided out after an odd step. Right: The dual chain on $\mathbb{Z}_2$ uses exactly two branches $\psi_0(x) = 2x$ and $\psi_1(x) = \frac{2x-1}{3}$ with equal probabilities, representing individual even and odd backward steps.

Lemma 4.1 (Geometric law of the 2-adic valuation). *For $a \geq 1$, the set of odd integers n with $\nu_2(3n+1) = a$ is exactly one residue class modulo 2^{a+1}, consisting of odd residues. Consequently, if n is drawn uniformly* *from the odd residues modulo 2^k, then* $\Pr[\nu_2(3n+1)=a]=2^{-a}$ *for every* $1\le a\le k-1$*.*

Proof. For odd n, $3n+1$ is even, so $\nu_2(3n+1)\ge 1$. For $a\ge 1$, $\nu_2(3n+1)=a$ iff $3n+1\equiv 2^a \pmod{2^{a+1}}$, i.e. $n\equiv 3^{-1}(2^a-1) \pmod{2^{a+1}}$, a single residue class modulo 2^{a+1} (3 is invertible modulo any power of 2); it consists of odd numbers because 2^a-1 and 3^{-1} are both odd. Among the 2^{k-1} odd residues modulo 2^k, this class contains exactly 2^{k-a-1} of them, giving probability $2^{k-a-1}/2^{k-1}=2^{-a}$. □

Definition 4.2 (Syracuse chain on $\mathbb{Z}_3$)**.** Since 2 is a unit in $\mathbb{Z}_3$, the maps

$$\varphi_a(x) \;=\; (3x+1)\,2^{-a}, \qquad a\ge 1,$$

are well defined on $\mathbb{Z}_3$. The *Syracuse chain* is the Markov kernel on $\mathbb{Z}_3$

$$P(x,\cdot) \;=\; \sum_{a\ge 1} 2^{-a}\,\delta_{\varphi_a(x)}.$$

By Lemma 4.1 this is the transfer model of the Syracuse map under which Tao constructs the Syracuse random variables [9].

Theorem 4.3 (Contraction on $\mathbb{Z}_3$)**.** ***Assumptions:*** *Let P be the Syracuse Markov chain on $\mathbb{Z}_3$ generated by the branches $\varphi_a(x)=(3x+1)2^{-a}$ with probabilities 2^{-a}. Let P_k denote its projection onto $\mathbb{Z}/3^k\mathbb{Z}$.*

Conclusion: *Each branch φ_a scales the 3-adic metric exactly by $1/3$, meaning $|\varphi_a(x)-\varphi_a(y)|_3 = \frac{1}{3}|x-y|_3$. Consequently, the Dobrushin–Wasserstein coefficient $\tau(P)$ is at most $1/3$, and the same bound holds for every projection P_k. In particular, P has a unique invariant measure π on $\mathbb{Z}_3$, distributions converge geometrically in W_1 at rate 3^{-n}, and* $\mathrm{spec}(U|_{\mathrm{Lip}(\mathbb{Z}_3)})\subseteq\{1\}\cup\{z:|z|\le 1/3\}$.

Interpretation: This result ensures that the probabilistic transfer model of the odd Collatz steps contracts geometrically in the Wasserstein metric. Although it does not prove pointwise convergence of individual trajectories (as $\mathbb{Z}_+$ is Haar-null in $\mathbb{Z}_3$), it guarantees that random distributions over trajectories converge uniquely and rapidly to the Syracuse measure, ruling out multiple competing stationary distributions.

Proof. For the branch identity: $\varphi_a(x)-\varphi_a(y)=3(x-y)2^{-a}$, and $|3|_3=\frac{1}{3}$, $|2^{-a}|_3=1$, so $|\varphi_a(x)-\varphi_a(y)|_3=\frac{1}{3}|x-y|_3$ exactly.

For $\tau(P)\le 1/3$, couple $P(x,\cdot)$ and $P(y,\cdot)$ *synchronously*: draw a single a with $\Pr[a]=2^{-a}$ and move both points along the same branch φ_a. This is a coupling of the two transition laws, of cost

$$\sum_{a\ge 1} 2^{-a}\,|\varphi_a(x)-\varphi_a(y)|_3 \;=\; \tfrac{1}{3}|x-y|_3,$$

so $W_1(P(x,\cdot),P(y,\cdot))\le\frac{1}{3}d(x,y)$ for all $x\ne y$.

For the projections: on $\mathbb{Z}/3^k\mathbb{Z}$ put $d_k(r,s) = 3^{-v_3(r-s)}$ for $r \neq s$ (well defined since $v_3(r-s) \leq k-1$). The projected kernel $P_k(r,\cdot)$ is the push-forward of $P(x,\cdot)$ under reduction modulo 3^k, for any lift x of r (the reduction of $\varphi_a(x)$ modulo 3^k depends only on $x \bmod 3^k$, indeed only on $x \bmod 3^{k-1}$). The same synchronous coupling projects to a coupling of $P_k(r,\cdot)$ and $P_k(s,\cdot)$; each coupled pair $(\varphi_a(x) \bmod 3^k,\ \varphi_a(y) \bmod 3^k)$ either coincides (cost 0, which happens when $v_3(x-y) = k-1$) or lies at distance $3^{-v_3(x-y)-1} = \frac{1}{3}d_k(r,s)$. Hence $W_1(P_k(r,\cdot), P_k(s,\cdot)) \leq \frac{1}{3}d_k(r,s)$ and $\tau(P_k) \leq \frac{1}{3}$.

The remaining assertions follow from Lemma 2.3 with $\tau = 1/3$. □

Remark 4.4. The bound $\tau(P_k) \leq 1/3$ is uniform in k, unlike the ℓ^2 spectral gap of the finite sections, which is trivial (Section 7.6: the projected chain loses all memory modulo 3^k in exactly k steps, so all eigenvalues beyond $\lambda_1 = 1$ vanish). The exact values of $\tau(P_k)$, computed in rational arithmetic in Section 7.7, are $\tau_2 = 5/21$, $\tau_3 = 455/1387$ and $\tau_4 = 7635497415/22906579627 \approx 0.33333206$, increasing towards the uniform bound: at finite levels, couplings across distinct branches still help, but their advantage vanishes as $k \to \infty$.

We now turn to the dual chain on $\mathbb{Z}_2$. The accelerated map T extends continuously to $\mathbb{Z}_2$ and preserves Haar measure; it is conjugate to the one-sided Bernoulli shift [10, 6]. Its transfer (backward) chain is driven by the two inverse branches of T on $\mathbb{Z}_2$:

$$\psi_0(x) = 2x \text{ (even step)}, \quad \psi_1(x) = \tfrac{2x-1}{3} \text{ (odd step)},$$

both well defined on $\mathbb{Z}_2$ since 3 is a unit in $\mathbb{Z}_2$.

Theorem 4.5 (Contraction and finite-time Haar averaging on $\mathbb{Z}_2$)**.** ***Assumptions:*** *Let Q be the Markov kernel on $\mathbb{Z}_2$ given by $Q(x,\cdot) = \frac{1}{2}\delta_{\psi_0(x)} + \frac{1}{2}\delta_{\psi_1(x)}$, with Koopman operator $Lf(x) = \frac{1}{2}f(2x) + \frac{1}{2}f\big(\frac{2x-1}{3}\big)$.*

Conclusion: *The Dobrushin–Wasserstein coefficient $\tau(Q)$ is exactly $1/2$. Haar measure on $\mathbb{Z}_2$ is the unique invariant measure of Q. Furthermore, if $f : \mathbb{Z}_2 \to \mathbb{R}$ depends only on $x \bmod 2^k$ (a level-k cylinder function), then $L^k f$ is identically equal to the exact Haar average $2^{-k}\sum_{r \bmod 2^k} f(r)$.*

Interpretation: On the 2-adic integers, the backward map acts essentially like a fair coin flip. The Wasserstein metric captures this perfectly: any observable that only depends on the first k digits of the state is completely forgotten (averaged out) after exactly k steps. This provides a clean, dimension-free geometric explanation for why parity vectors perfectly equidistribute.

Proof. Both branches scale the metric by exactly 1/2: $|\psi_0(x) - \psi_0(y)|_2 = |2|_2|x-y|_2 = \frac{1}{2}|x-y|_2$, and $|\psi_1(x) - \psi_1(y)|_2 = |2(x-y)/3|_2 = \frac{1}{2}|x-y|_2$ since $|3|_2 = 1$.

(1) The synchronous coupling (same branch on both points) gives cost exactly $\frac{1}{2}|x-y|_2$, so $W_1 \leq \frac{1}{2}|x-y|_2$. For the reverse inequality, note the

cross distances are maximal: $\psi_0(x) - \psi_1(y) = 2x - \frac{2y-1}{3} = \frac{6x-2y+1}{3}$ has odd numerator and unit denominator, so $|\psi_0(x) - \psi_1(y)|_2 = 1 \geq \frac{1}{2}|x-y|_2$ (as $|x-y|_2 \leq 1$), and symmetrically. Thus *every* coupling of the two two-point laws transports each unit of mass a distance at least $\frac{1}{2}|x-y|_2$, whence $W_1 \geq \frac{1}{2}|x-y|_2$.

(2) Uniqueness follows from (1) and Lemma 2.3. For invariance of Haar measure λ: ψ_0 is a homeomorphism of $\mathbb{Z}_2$ onto $2\mathbb{Z}_2$ (the even 2-adics) with $(\psi_0)_*\lambda = 2\,\lambda|_{2\mathbb{Z}_2}$ (it scales Haar mass by $|2|_2 = \frac{1}{2}$ onto a set of measure $\frac{1}{2}$, uniformly), and ψ_1 is a homeomorphism onto $1 + 2\mathbb{Z}_2$ (the odd 2-adics: $2x-1$ is odd and division by the unit 3 preserves 2-adic parity) with $(\psi_1)_*\lambda = 2\,\lambda|_{1+2\mathbb{Z}_2}$. Hence

$$\lambda Q \;=\; \tfrac{1}{2}(\psi_0)_*\lambda + \tfrac{1}{2}(\psi_1)_*\lambda \;=\; \lambda|_{2\mathbb{Z}_2} + \lambda|_{1+2\mathbb{Z}_2} \;=\; \lambda.$$

(3) For a word $u = (u_1, \dots, u_k) \in \{0,1\}^k$ write $\psi_u = \psi_{u_1} \circ \cdots \circ \psi_{u_k}$. Then

$$L^k f(x) \;=\; 2^{-k} \sum_{u \in \{0,1\}^k} f\big(\psi_u(x)\big).$$

Since each branch scales the metric by $\frac{1}{2}$, $|\psi_u(x) - \psi_u(y)|_2 = 2^{-k}|x-y|_2 \leq 2^{-k}$, so $\psi_u(x) \bmod 2^k$ does not depend on x; as f is a level-k cylinder function, each summand is a constant c_u. We claim the map $u \mapsto \psi_u(x) \bmod 2^k$ is a bijection onto $\mathbb{Z}/2^k\mathbb{Z}$: by induction on k, the images of ψ_0 and ψ_1 lie in the (disjoint) even and odd classes respectively, and within each class, ψ_i composed with the 2^{k-1} maps $\psi_{u'}$ ($u' \in \{0,1\}^{k-1}$) yields 2^{k-1} points that are pairwise distinct modulo 2^k, because ψ_i maps points distinct modulo 2^{k-1} to points distinct modulo 2^k (exact $\frac{1}{2}$-scaling of the metric). Hence $\{c_u\}_u$ enumerates $\{f(r) : r \bmod 2^k\}$ exactly once and $L^k f \equiv 2^{-k}\sum_r f(r) = \int f\,\mathrm{d}\lambda$. □

Remark 4.6. Theorem 4.5(3) is the functional-analytic reading of Terras's conjugacy with the Bernoulli shift [10]: the backward chain forgets the level-k observable in exactly k steps, i.e. mixing is not merely geometric but *finite-time* on cylinder observables. It is also the exact analogue of the finite-level rank collapse of the Syracuse chain (Section 7.6).

Remark 4.7 (Scope of the contraction theorems). Theorems 4.3 and 4.5 operate on spaces of *global densities* over the compact rings $\mathbb{Z}_3, \mathbb{Z}_2$. The set $\mathbb{Z}_+$ where the conjecture lives is Haar-null in both. Contraction of distributions in W_1 therefore does not, by itself, constrain individual orbits: in $\ell^1(\mathbb{Z}_+)$, every finite section of the pushforward operator we tested is nilpotent (spectrum $\{0\}$), and no elementary weight n^θ yields uniform contraction — the obstruction being, again, the residue classes $n \equiv -1 \bmod 2^t$ of Theorem 3.2 (Section 7.7). We state this limitation explicitly to avoid overclaiming: the contraction theorems quantify the measure-theoretic ingredient of "almost-all" results in the style of [10, 9]; they do not approach the pointwise conjecture.

5. Exact algorithms

This section describes the algorithms behind the verifications of Section 7 and proves the statements they rely on. Implementation details of the arithmetic are deferred to Section 6.

5.1. **Cycle enumeration via parity vectors.**

Proposition 5.1 (Affine form and cycle equation)**.** *Fix d odd and let $p = (p_0, \dots, p_{L-1}) \in \{0,1\}^L$ be the parity vector of the first L accelerated steps of n in the $3n + d$ system, with $k = \sum_i p_i$. Then*

$$T_d^L(n) \;=\; \frac{3^k n + b(p)}{2^L},$$

where $b(p) = b_L$ is given by the recurrence

$$b_0 = 0, \qquad b_{i+1} = \begin{cases} 3b_i + d\,2^i, & p_i = 1, \\ b_i, & p_i = 0. \end{cases}$$

Consequently n lies on a cycle of length L with parity vector p if and only if

$$n \;=\; \frac{b(p)}{2^L - 3^k}, \tag{2}$$

which for $L \geq 1$ has at most one solution ($2^L \neq 3^k$).

Proof. By induction on i: writing $k_i = p_0 + \cdots + p_{i-1}$, assume $T_d^i(n) = (3^{k_i} n + b_i)/2^i$ (true for $i = 0$). If $p_i = 0$, the next step halves: $T_d^{i+1}(n) = (3^{k_i} n + b_i)/2^{i+1}$ and $b_{i+1} = b_i$. If $p_i = 1$,

$$T_d^{i+1}(n) = \frac{3\,T_d^i(n) + d}{2} = \frac{3(3^{k_i} n + b_i)/2^i + d}{2} = \frac{3^{k_i+1} n + 3b_i + d\,2^i}{2^{i+1}},$$

matching the recurrence. The cycle condition $T_d^L(n) = n$ then reads $n(2^L - 3^k) = b(p)$. □

The enumeration algorithm tests (2) for every parity vector: every cycle other than the fixed point 0 contains an odd element, so it may be rotated to begin with $p_0 = 1$, and it suffices to sweep the 2^{L-1} vectors with $p_0 = 1$ for each $L \leq L_{\max}$. For each integral solution n, the algorithm *re-runs the actual orbit* and keeps n only if its true parity vector is p and $T_d^L(n) = n$ — eliminating algebraic solutions that do not realize their parity pattern. The output is therefore the complete list of cycles of T_d in $\mathbb{Z}$ of length at most $L_{\max}$: exact arithmetic replaces orbit simulation as the search mechanism.

5.2. **Diophantine cycle exclusion.**

Proposition 5.2 (Cycle inequality)**.** *Suppose every integer $2 \leq n \leq N$ converges to the trivial cycle under T, and let C be a nontrivial cycle of T*

in $\mathbb{Z}_+$ with k odd elements and total length L (in T-steps). Then all elements of C exceed N and

$$0 \;<\; L - k\alpha \;\le\; k\,\varepsilon(N), \qquad \varepsilon(N) := \log_2\Bigl(1 + \frac{1}{3N}\Bigr), \tag{3}$$

where $\alpha = \log_2 3$. In particular $\|k\alpha\| \le k\,\varepsilon(N)$.

Proof. If C contained an element $\le N$, that element would converge to the trivial cycle, contradicting nontriviality; so every element exceeds N. Let $x_1, \dots, x_k$ be the odd elements of C in cyclic order, and let 2^{a_i} be the power of two divided out between x_i and x_{i+1} (indices mod k), so $x_{i+1} = (3x_i+1)/2^{a_i}$ and $\sum_i a_i = L$. Multiplying around the cycle,

$$2^L \;=\; \prod_{i=1}^{k} \frac{3x_i + 1}{x_i} \;=\; \prod_{i=1}^{k}\Bigl(3 + \frac{1}{x_i}\Bigr).$$

Taking $\log_2$: $L - k\alpha = \sum_i \log_2\bigl(1 + \frac{1}{3x_i}\bigr)$, which is strictly positive and, since each $x_i > N$, at most $k\log_2(1 + \frac{1}{3N})$. As L is an integer, $\|k\alpha\| \le L - k\alpha \le k\varepsilon(N)$. ☐

The exclusion now follows from the classical theory of best rational approximation [4, Ch. I–II]: if p_i/q_i are the convergents of the (infinite, since α is irrational) continued fraction of α, then for $q_i \le k < q_{i+1}$,

$$\|k\alpha\| \;\ge\; \|q_i\alpha\| \;>\; \frac{1}{q_i + q_{i+1}},$$

so (3) forces $k > \bigl(\varepsilon(N)\,(q_i + q_{i+1})\bigr)^{-1}$ within that range. Sweeping consecutive convergent denominators yields the largest K such that every $k \le K$ is impossible; then $L > K\alpha$ bounds the cycle length, and since the L iterates of a cycle are distinct, also the number of its elements. This is the method of Eliahou [2] and Simons–de Weger [8]; our contribution is the certification scheme (Section 6) making every step exact. Numerical outcomes are in Section 7.3.

5.3. **The obstruction as a graph problem: Karp's algorithm.** Fix $j \ge 1$ and let $M = 2^j$. Define a weighted digraph G_j on the vertex set $\mathbb{Z}/M\mathbb{Z}$: each residue r has the two successors $T(r) \bmod M$ and $T(r+M) \bmod M$ (the images of the two lifts of r modulo $2M$, which determine T modulo M), and every edge leaving r carries the weight

$$c(r) \;=\; \begin{cases} \log_2 3 - 1, & r \text{ odd},\\ -1, & r \text{ even}.\end{cases}$$

The weight is the exact asymptotic logarithmic gain of the step on the class: for odd n, $\log_2 \frac{T(n)}{n} = \log_2\bigl(\frac{3}{2} + \frac{1}{2n}\bigr) \downarrow \log_2 3 - 1$ as $n \to \infty$, and for even n it is exactly -1.

Proposition 5.3 (Necessary feasibility relaxation)**.** *If a modular Lyapunov function of level j existed (Definition 3.1, extended to require decrease on even steps as well), then the difference constraints*

$$w(s) - w(r) \;\le\; -c(r) \qquad \textit{for every edge } r \to s \textit{ of } G_j \tag{4}$$

would be feasible. Conversely, (4) *is infeasible if and only if G_j contains a cycle of strictly positive total weight; and the maximum cycle mean of G_j equals* $\log_2 3 - 1 > 0$*, attained by the self-loop at the residue* $-1 \bmod M$*.*

Proof. Necessity. Let $r \to s$ be an edge of G_j with r odd, witnessed by the lift class $\{n \equiv r' \pmod{2M}\}$ (where $r' \in \{r, r+M\}$) whose members map into class s. For each positive n in that class, the assumed decrease gives $w(s) - w(r) < -\log_2 \frac{T(n)}{n} = -\log_2\bigl(\frac{3}{2} + \frac{1}{2n}\bigr)$. Letting $n \to \infty$ along the class, the right side increases to $-(\log_2 3 - 1) = -c(r)$, so $w(s) - w(r) \le -c(r)$. For r even the same argument gives $w(s) - w(r) \le 1 = -c(r)$ (with equality of the bound, since $\log_2 \frac{T(n)}{n} = -1$ exactly).

Feasibility criterion. If (4) holds and $r_0 \to r_1 \to \cdots \to r_\ell = r_0$ is a cycle, summing the constraints telescopes to $0 \le -\sum_i c(r_i)$, so every cycle has total weight ≤ 0. Conversely, suppose every cycle of G_j has total weight ≤ 0, i.e. has cost ≥ 0 for the edge costs $b(r \to s) := -c(r)$. Adjoin a super-source σ with a cost-0 edge to every vertex, and set $w(v)$ equal to the minimum cost of a walk from σ to v; this is well defined because walks cannot decrease cost by traversing cycles (all cycle costs are ≥ 0), so the minimum is attained by a simple path. By minimality, $w(s) \le w(r) + b(r \to s)$ for every edge, which is (4).

Maximum mean. Every edge weight is at most $\log_2 3 - 1$ (as $\log_2 3 - 1 > -1$), so every cycle has mean at most $\log_2 3 - 1$. The residue $r = M - 1 \equiv -1$ is odd, and its lift $2M - 1$ satisfies

$$T(2M-1) \;=\; \frac{3(2M-1)+1}{2} \;=\; 3M - 1 \;\equiv\; -1 \pmod{M},$$

so G_j has a self-loop at $-1 \bmod M$ of weight $\log_2 3 - 1$, attaining the bound. Its mean is positive precisely because $3 > 2$. ☐

Proposition 5.3 recasts Theorem 3.2 as the infeasibility of (4) and identifies the culprit cycle a priori. Karp's algorithm [3] computes the maximum cycle mean of G_j and an attaining cycle by dynamic programming in $O(|V||E|)$ time; running it for a range of j (Section 7.4) verifies that the *unique* optimal cycle is the predicted self-loop at $-1 \bmod 2^j$ — the computational fingerprint of the 2-adic fixed point.

5.4. **Exact transfer kernels and exact W_1 on ultrametric quotients.** The projected Syracuse kernel P_k on $\{r \bmod 3^k : 3 \nmid r\}$ is computed exactly: since 2 is a primitive root modulo 3^k, the multiplier $2^{-a} \bmod 3^k$ is periodic in a with period $\varphi(3^k) = 2 \cdot 3^{k-1}$, so the total transition mass onto each target residue is a finite geometric aggregate, a closed-form rational. Stationary measures are obtained by Gaussian elimination over $\mathbb{Q}$.

Wasserstein distances on the quotient are also exact, via the closed form on ultrametric spaces:

Lemma 5.4 (W_1 on $\mathbb{Z}/p^k\mathbb{Z}$)**.** *Let μ,ν be probability measures on $\mathbb{Z}/p^k\mathbb{Z}$ with the metric $d(x,y) = p^{-v_p(x-y)}$. For $1 \le \ell \le k$ let $m_\ell = \frac{1}{2}\sum_{c \bmod p^\ell}\big|\mu(c+p^\ell\mathbb{Z})-\nu(c+p^\ell\mathbb{Z})\big|$ be the total-variation discrepancy of the projections modulo p^ℓ. Then*

$$W_1(\mu,\nu) \;=\; \sum_{\ell=1}^{k-1}\big(p^{-(\ell-1)} - p^{-\ell}\big)\, m_\ell \;+\; p^{-(k-1)}\, m_k.$$

Proof. For $x \neq y$ the ultrametric distance decomposes along the tree of congruence cells: writing $[\, x \not\equiv y \ (p^\ell)\,]$ for the indicator,

$$d(x,y) \;=\; \sum_{\ell=1}^{k-1}\big(p^{-(\ell-1)} - p^{-\ell}\big)\,[\, x \not\equiv y \ (p^\ell)\,] \;+\; p^{-(k-1)}\,[\, x \neq y\,],$$

since the sum telescopes to $p^{-v_p(x-y)}$. Integrating against any coupling γ and using that the mass $\gamma\{x \not\equiv y \ (p^\ell)\}$ is at least the total-variation distance m_ℓ of the two projections modulo p^ℓ gives the lower bound $W_1 \ge \sum_\ell c_\ell m_\ell$ with the stated coefficients $c_\ell \ge 0$.

For the upper bound, construct a coupling greedily along the cell tree, from the finest level to the coarsest. First match, within each cell c modulo p^k, the common mass $\min\{\mu(c),\nu(c)\}$ at distance 0. The unmatched mass at level k has total m_k on each side; match as much of it as possible within cells modulo p^{k-1} (each such match travels distance $p^{-(k-1)}$), leaving unmatched totals m_{k-1} on each side — indeed, after all mass is matched within level-$(k-1)$ cells wherever possible, the surplus of μ over ν remaining in the cell c' modulo p^{k-1} is $(\mu(c') - \nu(c'))^+$, whose total is exactly m_{k-1}. Continuing upward, at each level $\ell = k-1,\dots,1$ the mass newly matched within level-ℓ cells is $m_{\ell+1} - m_\ell$, at distance $p^{-\ell}$ (mass conservation makes the within-cell surpluses independent of how the finer levels were matched), and finally m_1 is matched across level-1 cells at distance 1. The total cost is

$$\sum_{\ell=1}^{k} p^{-(\ell-1)}\big(m_\ell - m_{\ell-1}\big) \qquad (m_0 := 0),$$

which by Abel summation equals $\sum_{\ell=1}^{k-1}\big(p^{-(\ell-1)} - p^{-\ell}\big)m_\ell + p^{-(k-1)}m_k$, matching the lower bound. □

The contraction coefficient $\tau(P_k) = \max_{x\neq y} W_1(P_k(x,\cdot),P_k(y,\cdot))/d(x,y)$ is then a maximum of finitely many exact rationals. The values reported in Remark 4.4 were obtained this way.

6. Methodology: exact arithmetic

Because part of our evidence is computational, we specify the arithmetic model precisely; referees of computer-assisted results rightly demand this.

The guiding rule is: *no floating-point quantity participates in any certified claim.* Concretely:

(1) **Integers and rationals.** All integer arithmetic uses Python's built-in arbitrary-precision integers (no overflow is possible; numbers such as 3^q with $q \approx 10^5$ are handled exactly). All rational arithmetic (transfer matrices, stationary measures, Wasserstein distances, the coefficients τ_k, the cycle equation (2)) uses exact fractions from the standard `fractions` module. Reported values such as $\tau_3 = 455/1387$ are exact outputs, with decimal expansions given only as display approximations.

(2) **The constant $\log_2 3$.** The only irrational constant in the toolkit enters at two places, each handled without trusting floating point:
   - *Convergents* (Section 5.2). The partial quotients of $\alpha = \log_2 3$ are extracted from a 120-digit decimal approximation (`decimal` module, correctly rounded to the working precision), and every convergent p/q with $q \leq 10^5$ is then *certified exactly*: the predicted side of the alternation $p/q \gtrless \alpha$ is equivalent to the integer comparison $2^p \gtrless 3^q$, which is checked in exact integer arithmetic. The certification failing would raise an error; it never does within the range used by our bounds.
   - *The slack* $\varepsilon(N)$ (Proposition 5.2). The algorithm replaces $\varepsilon(N)$ by a certified rational *upper* bound (the 120-digit value rounded up by one unit in the last place). Any overestimate of ε only weakens the resulting bound K, never its validity.

(3) **Where floats do appear.** Karp's algorithm (Section 5.3) runs its dynamic program on IEEE double weights for speed. Its output is not trusted as such: the value of the maximum cycle mean is *proved* to be $\log_2 3 - 1$ (Proposition 5.3), and the run is accepted only if the extracted optimal cycle consists solely of odd residues — a symbolic condition checked exactly, under which the cycle's mean is $\log_2 3 - 1$ by construction. The float value agrees with $\log_2 3 - 1$ to $< 2 \cdot 10^{-15}$ in all runs, as expected. Likewise, the power-iteration estimate of the second eigenvalue of P_k is a numerical diagnostic only; the corresponding certified statement is the exact rank collapse $P_k^k = \mathbf{1}\pi_k$ verified in rational arithmetic (Section 7.6).

(4) **Determinism.** All algorithms are deterministic; there is no Monte Carlo component anywhere in the toolkit. Re-running the reproduction script yields bit-identical results on any platform (the test suite runs on CPython 3.9–3.13 on Linux and macOS in continuous integration).

7. Computational verification

Every number in this section is regenerated and checked by the reproduction script `reproduce_paper_results.py`; the labels R1–R8 refer to its sections. We emphasize (cf. Section 1.4) that items 7.1–7.5 are *verifications with explicit finite limits*, not theorems, while the exact rational outputs in 7.6–7.7 are finite-level instances of Theorems 4.3 and 4.5.

Claim	Verified by	Exact?
Cycle enumeration	Rational arithmetic / cycle equation (2)	Yes
Cycle exclusion bounds	Continued fraction certificates	Yes
Obstruction loop at -1	Karp's dynamic program / symbolic verification	Yes
W_1 Contraction coeffs	Finite exact transfer kernels	Yes

TABLE 1. Summary of computational verifications, demonstrating the exact reproducibility of all programmatic claims.

7.1. **Convergence sieve (R1).** Every $1 \leq n \leq 2 \cdot 10^5$ reaches the trivial cycle under T (ascending sieve: each n is iterated only until it drops below itself). This is far below the published record 2^{68} [1] and is included solely as a self-contained, re-runnable hypothesis for 7.3. The largest stopping time in the range (accelerated steps until the orbit first drops below its seed) is attained at $n = 35{,}655$, with 135 steps; the largest excursion is attained at $n = 159{,}487$, whose orbit peaks at 8,601,188,876.

7.2. **Cycle enumeration (R2).** The enumeration of Section 5.1, swept over all parity vectors of length ≤ 14:

System	Cycles found (by least-$|\cdot|$ element)	Status
$3n+1$, in $\mathbb{Z}$	$\{1, 2\}$; $\{-1\}$; $\{-5, -7, -10\}$; cycle of -17 (len. 11)	complete, $L \leq 14$
$3n-1$, in $\mathbb{Z}_+$	$\{1\}$; $\{5, 7, 10\}$; cycle of 17 (len. 11) (†)	validation
$3n+5$, in $\mathbb{Z}_+$	$\{1, 4, 2\}$; $\{5, 10\}$; $\{19, 31, 49, 76, 38\}$; $\{23, 37, 58, 29, 46\}$	validation

(†) For $3n - 1$ the listed cycles are the positive ones; by the conjugation $n \mapsto -n$ they are exactly the negative cycles of $3n + 1$ (verified exactly: $\varphi(x) = -x$ satisfies $\varphi \circ T_1 = T_{-1} \circ \varphi$, R8).

The enumeration over $\mathbb{Z}$ is *complete* up to the swept length: on the positives of $3n + 1$, only the trivial cycle exists with $L \leq 14$; the three negative cycles are rediscovered by the algorithm, validating the detector on known ground truth.

7.3. **Diophantine exclusion (R3).** Proposition 5.2 with the certified convergents of α:

Hypothesis (all $n \leq N$ converge)	odd steps k of any cycle	elements
$N = 2 \cdot 10^5$ (verified here, R1)	$k > 428$	> 676
$N = 2^{71}$ (cf. [1])	$k > 65{,}470{,}613{,}320$	$> 103{,}443{,}569{,}045$

The second row uses the verification limit reported by the ongoing project of [1]; with the published 2^{68} the bound is of the same order. These reproduce the method of [2, 8].

7.4. **Locating the obstruction (R4).** Karp's algorithm on G_j for $j = 5, \ldots, 10$: in every run the maximum mean equals $\log_2 3 - 1 = 0.5849625\ldots$ (float agreement $< 2 \cdot 10^{-15}$; value certified symbolically per Section 6), and the optimal cycle returned is *unique and equal to the self-loop at* $-1 \bmod 2^j$ predicted by Proposition 5.3 — the finite-level fingerprint of Theorem 3.2.

7.5. **Inverse tree of the trivial cycle (R5).** The conjecture is equivalent to the statement that the preimage tree of 1 under T (edges $m \to 2m$ and, when integral and odd, $m \to (2m-1)/3$) covers $\mathbb{Z}_+$. Exact BFS gives: the minimum depth at which the tree rooted at 1 covers all of $\{1, \ldots, 1000\}$ is 113 (the depth of $n = 871$, equivalently the maximum total stopping time in the range); and at level 30 the new nodes span exactly the interval of values $[123,\ 2^{31}]$ (minimum and maximum new node). Growth per level is $\approx 4/3$, consistent with the branching heuristic; Krasikov–Lagarias prove the reached set below X has cardinality $\geq X^{0.84}$ [5].

7.6. **Finite-level Syracuse spectra (R6).** For the projected kernel P_k (exact rationals, Section 5.4): the uniform measure is *not* stationary; the exact stationary measure modulo 3 is $\pi(1) = 1/3$, $\pi(2) = 2/3$ — Syracuse iterates fall on 2 mod 3 twice as often as on 1 mod 3, an exact consequence of $(3n+1)/2^a \equiv (-1)^a \pmod 3$ with $\Pr[a \text{ odd}] = 2/3$ under Lemma 4.1. Moreover $P_k^k = \mathbf{1}\pi_k$ *exactly* (rank collapse verified in rational arithmetic for the tested k): the chain loses all memory modulo 3^k in exactly k steps, so the ℓ^2 spectrum beyond $\lambda_1 = 1$ is $\{0\}$ — which is why the surviving contraction structure must be sought in the Wasserstein geometry (Theorem 4.3), not in finite ℓ^2 sections.

7.7. **Exact Wasserstein coefficients (R7).** Lemma 5.4 applied to the exact kernels:

Level	exact τ_k	decimal
3^2	$5/21$	$0.238095\ldots$
3^3	$455/1387$	$0.328046\ldots$
3^4	$7635497415/22906579627$	$0.33333206\ldots$
$\mathbb{Z}_3$ (Theorem 4.3)	$\leq 1/3$ (uniform bound)	$0.\overline{3}$
2^6 on $\mathbb{Z}_2$ (Theorem 4.5)	$1/2$ (exact)	0.5

The coefficients increase monotonically towards the uniform bound 1/3, and each branch-contraction identity, the projective coherence of the family π_k, and the finite-time Haar averaging $L^k f = \int f \, d\lambda$ of Theorem 4.5(3) are verified in exact arithmetic at the tested levels. On the other side of Remark 4.7, the finite sections of the pushforward operator on $[1, N] \subset \mathbb{Z}_+$ are exactly nilpotent (acyclic transition structure) for the tested $N \leq 10^5$, and the weight n^θ fails to contract uniformly on the classes $n \equiv -1 \bmod 2^t$, as predicted by Theorem 3.2.

7.8. **Structural cross-checks (R8).** Terras's bijection (parity vector bijective on $\mathbb{Z}/2^{14}\mathbb{Z}$) verified exactly; the affine conjugation $\varphi(x) = -x$ between $3n+1$ and $3n-1$, and the embedding $\varphi(x) = 5x$ of $3n+1$ into $3n+5$, verified exactly on the tested ranges.

8. Discussion

Primary contribution and implications. The primary contribution of this paper is the complete obstruction theorem (Theorem 3.2), which closes off an entire natural family of proof strategies for the Collatz conjecture. We have shown that no potential depending on finitely many residue classes modulo a power of two can convert the average logarithmic descent of T into a monotone certificate on $\mathbb{Z}_+$. The identified obstruction is the "right" one because it arises from a fundamental topological feature: the 2-adic fixed point -1. This single point is invisible to finite modular arithmetic but arbitrarily well approximated by positive integers, permanently breaking the 2-adic symmetry between $\mathbb{Z}_+$ and the negatives in any finite projection.

Viable Lyapunov approaches. Is the modular family sufficiently broad that its failure represents a major barrier? Yes, because it encompasses all bounds derived purely from finite parity-vector matching. However, this does not mean all Lyapunov approaches are doomed. Any successful argument must break the symmetry between $\mathbb{Z}_+$ and $\mathbb{Z}_-$. Viable approaches might include:

- **Nonlocal corrections:** Potentials that depend on the full, infinite forward trajectory, effectively using the exact geometry of the attractor rather than finite approximations.
- **Unbounded or aperiodic weights:** Corrections that grow with n rather than remaining strictly periodic.
- **Probabilistic Lyapunov functions:** Functions that decrease in expectation over multiple steps, circumventing deterministic block-decrease obstructions.

Wasserstein contraction and prior theory. Our secondary contribution (Theorems 4.3 and 4.5) identifies the exact norm in which the transfer structure of the map contracts uniformly (Wasserstein over the compact p-adic rings, with coefficients $1/3$ and $1/2$). One might ask if these results already follow from existing transfer-operator theory. While rapid mixing is well known [10, 12], previous formulations often rely on functional-analytic bounds or lack explicit, dimension-free constants. Our bounds are exact and purely geometric, providing a much simpler mechanism for finite-time Haar averaging that contrasts sharply with the trivial ℓ^2 spectra of all finite sections.

Extensions and related systems. The same structural mechanism applies to generalized $3n + d$ maps: any p-adic fixed point or cycle that projects a positive logarithmic gain into finite quotients will obstruct modular Lyapunov functions in those systems. Furthermore, one might investigate whether similar obstructions appear modulo odd primes. While our proof relies heavily on the dyadic valuation (which directly determines the sequence of operations), analogous p-adic cycles could play a similar role in generalized $pn+d$ maps, though the transition logic is inherently dependent on the modulus defining the map.

Computational independence. Finally, we emphasize the reproducibility of our computational verifications. All computational claims are based on elementary, exact integer arithmetic and certified bounds. Are these certificates independent enough that another implementation would reproduce them? Yes. The code contains no heuristics, no floating-point arithmetic in certified bounds, and uses standard, unambiguous mathematical identities (such as the cycle equation (2) and the exact rational Wasserstein formula). Any independent software library replicating these algebraic steps will inevitably reach the exact same rational coefficients and cycle bounds.

Data and code availability

The Python package `collatz`, the test suite, the reproduction script `reproduce_paper_results.py`, and this manuscript are available in the accompanying repository under the MIT license (code) and CC BY 4.0 (text). All results reproduce with CPython ≥ 3.9 with no third-party dependencies.

References

1. David Barina, *Convergence verification of the Collatz problem*, The Journal of Supercomputing **77** (2021), no. 3, 2681–2688.
2. Shalom Eliahou, *The* $3x+1$ *problem: new lower bounds on nontrivial cycle lengths*, Discrete Mathematics **118** (1993), no. 1–3, 45–56.
3. Richard M. Karp, *A characterization of the minimum cycle mean in a digraph*, Discrete Mathematics **23** (1978), no. 3, 309–311.
4. Aleksandr Ya. Khinchin, *Continued fractions*, The University of Chicago Press, Chicago, 1964, Reprinted by Dover, 1997.
5. Ilia Krasikov and Jeffrey C. Lagarias, *Bounds for the* $3x+1$ *problem using difference inequalities*, Acta Arithmetica **109** (2003), no. 3, 237–258.
6. Jeffrey C. Lagarias, *The* $3x+1$ *problem and its generalizations*, The American Mathematical Monthly **92** (1985), no. 1, 3–23.
7. Jeffrey C. Lagarias (ed.), *The ultimate challenge: The* $3x+1$ *problem*, American Mathematical Society, Providence, RI, 2010.
8. John Simons and Benne de Weger, *Theoretical and computational bounds for* m*-cycles of the* $3n+1$ *problem*, Acta Arithmetica **117** (2005), no. 1, 51–70.
9. Terence Tao, *Almost all orbits of the Collatz map attain almost bounded values*, Forum of Mathematics, Pi **10** (2022), e12.
10. Riho Terras, *A stopping time problem on the positive integers*, Acta Arithmetica **30** (1976), no. 3, 241–252.
11. Cédric Villani, *Optimal transport: Old and new*, Grundlehren der mathematischen Wissenschaften, vol. 338, Springer, Berlin, 2009.

12. Günther J. Wirsching, *The dynamical system generated by the* $3n+1$ *function*, Lecture Notes in Mathematics, vol. 1681, Springer, Berlin, 1998.

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